

# **LEARNX E LEARNING PLATFORM**

**A PROJECT REPORT**

Submitted by

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**CUAYMCA006**

In Partial Fulfilment of the Requirements for the Degree of  
**MASTER OF COMPUTER APPLICATION (MCA)**



**CENTRE FOR COMPUTER SCIENCE AND INFORMATION  
TECHNOLOGY (CCSIT)  
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## **DECLARATION**

I **FIDHA K**, hereby declare that the project work entitled “**LEARNX**” is original work done by me, under the guidance of **Ms. SONA RAVINDRAN**, Assistant Professor, Centre for Computer Science and Information Technology, University of Calicut. This project report is submitted in partial fulfilment of the requirement for the award of the degree of Master of Computer Application during the period of study at University of Calicut.

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# CERTIFICATE

Certified that this is a bonafide record of the project work entitled “**LEARNX**” submitted by **FIDHA K (Reg No: CUAYMCA006)** in partial fulfilment of the requirement for the award of the degree of **MASTER OF COMPUTER APPLICATION** from **University of Calicut**.

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Certified that the candidate was examined by us in the Project Viva Voce Examination held on .....and her Register Number is .....

Examiners

.....

.....

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# INDEX

| Contents                                 | Page No. |
|------------------------------------------|----------|
| 1.INTRODUCTON                            | 10       |
| INTRODUCTION                             | 11       |
| 2.PROBLEM DEFINITION AND METHODOLOGY     | 12       |
| PROBLEM DEFINITION                       | 13       |
| OBJECTIVES                               | 13       |
| MOTIVATION                               | 14       |
| METHODOLOGY                              | 14       |
| SCOPE                                    | 15       |
| 3.REQUIREMENT ANALYSIS AND SPECIFICATION | 17       |
| EXISTING SYSTEM                          | 18       |
| PROPOSED SYSTEM                          | 18       |
| FEASIBILITY STUDY                        | 22       |
| 4.SYSTEM DESIGN                          | 26       |
| USERS OF THE SYSTEM                      | 27       |
| MODULARITY CRITERIA                      | 28       |
| ARCHITECTURE DESIGN                      | 29       |
| DATA FLOW DIAGRAM                        | 32       |
| DATABASE TABLES                          | 36       |
| 5.SYSTEM DEVELOPMENT                     | 43       |
| SYSTEM DEVELOPMENT                       | 45       |
| 6.SYSTEM TESTING AND IMPLEMENTATION      | 47       |
| TESTING TECHNOLOGIES                     | 48       |
| SYSTEM IMPLEMENTATION                    | 51       |
| 7.SYSTEM MAINTENANCE                     | 52       |
| SYSTEM MAINTENANCE                       | 53       |
| 8.CONCLUSION                             | 54       |
| CONCLUSION                               | 55       |
| SCOPE AND FUTURE ENHANCEMENT             | 56       |
| 9.BIBLIOGRAPHY                           | 57       |
| BIBLIOGRAPHY                             | 58       |
| 10.APPENDIX                              | 59       |
| SCREENSHOTS                              | 60       |

## **ABSTRACT**

LEARNX is an innovative e-learning platform designed to address the critical limitations of traditional online education by providing personalized, high-quality instruction for all categories of learners. Existing systems fail to integrate real-time interaction with intelligent assessment, leaving students with pre-recorded content, manual instructor coordination, and no automated diagnostic feedback. LEARNX overcomes these challenges by combining structured content delivery, intelligent instructor matching based on individual learning requirements and availability, live interactive sessions, and AI-driven diagnostic assessment. By automating attendance tracking, progress monitoring, and weakness identification, the platform ensures that knowledge gaps are systematically addressed. This integrated approach eliminates inefficient learning experiences, reduces wasted time, and promotes complete subject mastery, establishing LEARNX as a comprehensive solution for modern knowledge hunting.

## LIST OF FIGURES

| SL NO. | FIGURES                 | PAGE NO |
|--------|-------------------------|---------|
| 1      | Figure 1(level-0 dfd)   | 29      |
| 2      | Figure 2(level-1.1 dfd) | 29      |
| 3      | Figure 3(level-2.1 dfd) | 30      |

## LIST OF TABLES

| SL NO. | TABLES              | PAGE NO. |
|--------|---------------------|----------|
| 1      | Table 1(FileRecord) | 31       |



# **LEARNX E LEARNING PLATFORM**

# INTRODUCTION

## INTRODUCTION

LEARNX is a e learning platform that can be used for all category people for their knowledge hunting. In today's rapidly evolving educational landscape, learners face significant challenges in accessing personalized, high-quality instruction tailored to their individual needs. Traditional online learning platforms offer either pre-recorded content without real-time interaction or live tutoring sessions that require tedious manual coordination. Students struggle to find appropriate instructors based on their specific learning requirements, availability constraints, and subject preferences. Furthermore, existing systems lack intelligent mechanisms to assess student performance beyond surface-level evaluation. When learners complete sessions, there is no automated diagnostic analysis to identify their specific areas of weakness, leaving knowledge gaps unaddressed. Attendance tracking and progress monitoring remain manual processes, providing students with no consolidated view of their learning journey. These limitations result in inefficient learning experiences, wasted time, and incomplete mastery of subjects, highlighting the critical need for an integrated platform that combines structured content delivery, intelligent instructor matching, live interaction, and AI-driven diagnostic assessment.

## **PROBLEM DEFINITION AND METHODOLOGY**

## **PROBLEM DEFINITION AND SYSTEM STUDY**

### **PROBLEM DEFINITION**

In today's rapidly evolving educational landscape, learners face significant challenges in accessing personalized, high-quality instruction tailored to their individual needs. Traditional online learning platforms offer either pre-recorded content without real-time interaction or live tutoring sessions that require tedious manual coordination. Students struggle to find appropriate instructors based on their specific learning requirements, availability constraints, and subject preferences. Furthermore, existing systems lack intelligent mechanisms to assess student performance beyond surface-level evaluation. When learners complete sessions, there is no automated diagnostic analysis to identify their specific areas of weakness, leaving knowledge gaps unaddressed. Attendance tracking and progress monitoring remain manual processes, providing students with no consolidated view of their learning journey. These limitations result in inefficient learning experiences, wasted time, and incomplete mastery of subjects, highlighting the critical need for an integrated platform that combines structured content delivery, intelligent instructor matching, live interaction, and AI-driven diagnostic assessment.

### **OBJECTIVES**

The primary objective of this project is to design and develop a comprehensive global elearning platform that delivers personalized, mastery-based education across all subject domains through an integrated technological solution. The system aims to create a hierarchical knowledge taxonomy that organizes educational content from broad fields down to specific sub-topics, enabling learners to precisely navigate and select their desired areas of study with clarity and structure. A key objective is to implement a dynamic, user-driven scheduling system with dual instructor matching mechanisms, allowing students to either directly assign teachers or receive AI-generated suggestions based on expertise, availability, and learner preferences, thereby eliminating the inefficiency of manual coordination. The platform further seeks to establish live, interactive virtual classrooms that facilitate real-time instruction, doubt clarification, and personalized guidance between students and educators, replicating the benefits of face-to-face learning in a digital environment. Another critical objective is to design an AI-proctored examination system that not only evaluates student performance but also conducts diagnostic analytics to pinpoint specific areas of weakness for targeted improvement, moving beyond surface-level assessment to drive genuine concept mastery. Additionally, the project aims to integrate comprehensive tracking functionalities that automatically record attendance, participation duration, and performance metrics, providing users with detailed insights into their learning progress over time. Finally, the system is to be built using a scalable modern technology stack including Flutter for cross-platform frontend, Node.js for backend logic, and MongoDB for flexible data management, ensuring reliability, performance, and

accessibility across devices for learners worldwide.

## **MOTIVATION**

The motivation for developing the LEARNX stems from the critical need to help creators, businesses, and individuals protect and manage their digital assets by maximizing data sovereignty and minimizing operational dependencies on centralized tech giants. In competitive digital markets, time and money spent dealing with arbitrary cloud service takedowns, opaque security configurations, and hidden multi-platform transaction fees directly erode an entity's independence and bottom line. Our system is driven by the goal of eliminating these structural inefficiencies.

By creating a platform that integrates a fast, Web3-optimized storage system with MetaMask wallet authentication and real-time smart contract state updates, the system directly addresses data loss from server downtime, compromised credentials, and unauthorized third-party intervention. It provides a distinct competitive edge by centralizing decentralized identity validation, peer-to-peer file access whitelists, and marketplace distribution data for easy access.

Ultimately, the core motivation is to empower the asset owner by replacing corporate guesswork with immutable blockchain data. The system provides an immediate, clear view of the real storage status, permission logs, and asset monetization values, allowing for instant, informed decision-making regarding digital rights distribution, file pricing, and repository structure, leading to improved asset security and sustainable digital growth.

## **METHODOLOGY**

The development of the LearnX was executed following an Agile Methodology due to its iterative and highly flexible approach. This strategy ensured the project was continuously aligned with real-world personalized learning requirements and allowed for regular evaluation and integration of key educational and AI-driven features.

### **1. Incremental Development**

The system was built in functional, discrete modules, starting with foundational components like user authentication (email and social login), role-based dashboards for learners and instructors, and basic session scheduling. Progressing to complex modules such as the AI-driven diagnostic assessment engine, live video streaming integration, intelligent instructor-learner matching algorithm, and automated remedial content recommendation system.

## 2. User-Centric Design

Continuous consideration of learner and instructor workflows influenced the design of critical features, such as the highly intuitive personalized dashboard layout, seamless navigation between live sessions and content repository, real-time attendance tracking, and the effortless generation of AI diagnostic reports for immediate feedback and improvement.

## 3. Flexibility and Adaptability

The Agile framework provided the essential flexibility to seamlessly accommodate evolving educational requirements, such as refining the instructor verification and onboarding process, optimizing the matching algorithm for better learner-instructor compatibility, enhancing the diagnostic report structure based on educator feedback, and adding new filtering and sorting options to the content repository and session marketplace.

## 4. Focus on Deliverables and Continuous Improvement

Essential features like user registration, role-based access control, live session hosting and joining, basic content repository access, and automated attendance tracking were prioritized for early completion. The methodology also inherently allows for future expansion, such as integrating blockchain-based credentialing and certification, developing native mobile applications with offline learning mode, implementing gamification and peer learning engines, or adding NLP-powered intelligent chatbots for 24/7 academic support, ensuring the system remains scalable and aligned with evolving educational technology standards.

## Why Agile?

Agile was the ideal methodology as it enabled regular deployment of functional frontend and backend components, maintained a strong focus on immediate user experience constraints (responsiveness, low latency, and accessibility), and facilitated continuous improvement based on real-time feedback from learners and instructors. This resulted in a robust, scalable, and market-ready LearnX intelligent e-learning environment.

## **SCOPE**

The scope of LEARNX encompasses the design and development of an integrated e-learning platform that caters to learners across all categories, including students, working professionals, and lifelong learners. The platform will feature an intelligent instructor-learner matching system based on subject preferences, availability, and learning styles. It will support live interactive sessions with real-time communication capabilities, alongside a structured repository of pre-recorded content. A key component of the scope is the implementation of an AI-driven diagnostic assessment engine that automatically analyzes learner performance after each session, identifies specific knowledge gaps, and generates personalized remediation recommendations. Additionally, the platform will automate attendance tracking and provide a consolidated dashboard for learners to monitor their progress, session history, and competency levels. The system will not include offline delivery mechanisms, physical tutoring infrastructure, or certification generation, as these fall outside the current development phase. By focusing on intelligent matching, live interaction, and automated diagnostics, LEARNX aims to eliminate manual coordination and ineffective learning practices within a fully online environment.



## **REQUIREMENT ANALYSIS AND SPECIFICATION**

# **REQUIREMENT ANALYSIS AND SPECIFICATION**

## **EXISTING SYSTEM**

In the current online education landscape, learners primarily engage with platforms offering either pre-recorded content or live tutoring sessions. Most existing systems provide structured courses with video lectures and automated quizzes but lack real-time instructor interaction and personalized adaptation. Their assessment mechanisms are limited to surface-level multiple-choice questions without deep diagnostic analysis of student weaknesses. If a learner struggles with a concept, the system fails to identify or address that gap. Live tutoring platforms require users to manually browse tutor profiles and coordinate session timings through tedious back-and-forth communication, with no intelligent scheduling or automated matching. These platforms lack structured knowledge hierarchies, resulting in ad-hoc sessions without coherent progression. Attendance tracking and participation metrics are either absent or require manual entry. Another major drawback is that students must follow predetermined course paths without adaptation to individual learning pace or comprehension level. The systems do not understand the actual meaning of student queries or performance patterns. If a learner needs help with a specific topic, they cannot receive targeted recommendations or automatically matched instructors based on their weaknesses. Due to these limitations—static content, no diagnostic assessment, manual scheduling, absent progress tracking, and lack of personalization—traditional e-learning systems fail to deliver a truly effective educational experience, highlighting the need for an integrated intelligent platform.

## **PROPOSED SYSTEM**

The proposed system is an intelligent, AI-driven e-learning platform designed to deliver personalized, mastery-based education across all subject domains. It integrates structured knowledge organization, intelligent scheduling, live interactive instruction, and automated diagnostic assessment into a unified solution. The application organizes knowledge through a hierarchical taxonomy, allowing learners to navigate from broad fields down to specific sub-topics (e.g., Fashion > Dressmaking > Stitching). This structured approach enables precise selection of study areas and ensures logical learning progression. Upon logging in, students specify their desired field, preferred session timing, and duration. The system then intelligently matches them with available instructors through a dual mechanism: direct teacher assignment by the user or AI-generated suggestions based on instructor expertise, availability, and student preferences. Live, interactive virtual classrooms serve as the primary instructional medium, enabling real-time communication and personalized guidance. Following each session, an AI-proctored examination evaluates performance and performs diagnostic analytics to pinpoint specific areas of weakness for targeted improvement. These insights inform future session recommendations. The platform integrates comprehensive tracking functionalities, automatically recording attendance,

participation duration, and performance metrics. Students monitor progress over time while instructors gain visibility into class engagement. Built with Flutter for cross-platform experience, Node.js for backend logic, and MongoDB for flexible data management, the proposed system delivers a scalable, data-informed learning ecosystem addressing critical limitations of existing solutions.

## **REQUIREMENT SPECIFICATION**

### **A) FUNCTIONAL REQUIREMENTS**

- A functional requirement specifies what a system, application, or process should do. It defines the features, behaviors, and tasks the system must perform to fulfill its intended purpose. These requirements focus on the core functionality of the system, described as inputs, outputs, and processing steps.
- - Role-Based Authentication: The system must allow learners, instructors, and administrators to securely register and log in using email-based authentication or social login options. It must enforce role-based access restrictions, ensuring that only instructors can create and manage live sessions, only learners can enroll and access personalized content, and only administrators can oversee platform analytics and user management.
- - Personalized Learner Dashboard: The system shall provide each learner with a consolidated dashboard displaying upcoming live sessions, completed session history, attendance records, AI-generated weakness reports, and personalized recommendations for remedial content based on diagnostic assessments.
- - Intelligent Instructor-Learner Matching: The system must automatically match learners with appropriate instructors based on subject preferences, availability constraints, learning style compatibility, and past performance data. The matching algorithm shall prioritize instructor expertise relevancy and time slot alignment.
- - Live Interactive Session Management: The system must enable instructors to schedule, host, and record live video sessions with real-time chat, screen sharing, and whiteboard capabilities. Learners must be able to join scheduled sessions, participate interactively, and access recorded sessions for future review.

- - AI-Driven Diagnostic Assessment Engine: The system must automatically analyze learner performance after each completed session, identify specific knowledge gaps and weak areas, generate detailed diagnostic reports, and recommend targeted remedial content or practice exercises without requiring manual instructor intervention.
- - Automated Attendance Tracking: The system must automatically record learner attendance for each live session based on join time, duration of participation, and engagement metrics. Attendance records must be consolidated into the learner's progress dashboard and made available for instructor review.
- - Structured Content Repository: The system must allow instructors to upload and organize pre-recorded video lectures, reading materials, quizzes, and assignments into categorized subject folders. Learners must be able to access this content on demand, with progress tracking across completed materials.
- - Progress Monitoring and Analytics: The system must maintain a comprehensive learner progress ledger, tracking session completion rates, assessment scores, time spent on remedial content, and overall mastery levels across subjects. Visual progress charts and trend analysis must be presented to both learners and instructors.
- - Remedial Content Recommendation: The system must automatically generate and push personalized remedial learning paths to learners based on AI-identified weak areas. This includes suggesting specific video modules, practice quizzes, or one-on-one revision sessions with instructors.

## **B) NON-FUNCTIONAL REQUIREMENTS**

- A non-functional requirement specifies the quality attributes, constraints, and standards that the system must adhere to. These requirements define how the system

performs its functions, focusing on aspects like performance, scalability, usability, and security.

- - Usability: The web application must feature a clean, intuitive, and distraction-free interface modeled after modern learning management systems (such as Google Classroom or Coursera). Learners of all age groups and technical backgrounds must be able to navigate the platform, join live sessions, access diagnostic reports, and track progress without requiring extensive training or documentation.
- - Performance: The system must be highly responsive, ensuring that dashboard loading, session joining, and report generation occur with minimal latency (under 3 seconds for standard operations). Live video streaming must maintain low buffering and high synchronization, while AI diagnostic reports should be generated within 5 seconds of session completion.
- - Scalability: The application architecture must be designed to accommodate a growing number of concurrent users, supporting at least 1,000 simultaneous live session participants without degradation in video quality or response times. The system must maintain modularity between the video streaming layer, AI processing engine, and database backend to allow independent scaling of high-demand components.
- - Cross-Device Accessibility: The frontend application must be fully responsive and function correctly across modern web browsers (Chrome, Firefox, Safari, Edge) as well as mobile devices (iOS and Android). Learners must be able to join live sessions, view recorded content, and access diagnostic reports from smartphones, tablets, laptops, and desktop computers.
- - Security: The system must protect user data including personal information, session recordings, and diagnostic reports through encrypted storage and secure HTTPS transmission. Role-based access controls must be strictly enforced to prevent unauthorized access to instructor-only features or cross-learner data. All API connections to third-party services (video streaming, AI processing) must use securely stored environment variables and authentication tokens.

- - Reliability: The system must guarantee 99.5% uptime for live session hosting and content access. Automated backups of learner progress data, session recordings, and diagnostic reports must be performed daily to prevent data loss. In case of network interruptions during live sessions, the system must automatically attempt reconnection and preserve partial attendance records.
- - Accessibility Compliance: The platform must comply with WCAG 2.1 Level AA accessibility standards, including support for screen readers, keyboard navigation, sufficient color contrast, and captions for all video content to ensure equal access for learners with disabilities.

## **C) ENVIRONMENT DETAILS**

### **i) HARDWARE SPECIFICATIONS**

The selection of hardware is very important in the existence and proper working of any of the software. When selecting hardware, the size and capacity requirements are also important. The hardware must suit all application developments, especially for running a highly interactive, real-time decentralized cloud storage and web application.

- PROCESSOR: Intel Core i5 / AMD Ryzen 5 (or newer equivalent) to support local blockchain simulation, smart contract compilation, and cryptographic hashing.
- RAM: 8GB DDR4 RAM or Above (16GB recommended for smoothly running concurrent Hardhat nodes, Node.js local backend pipelines, and browser instances).
- STORAGE: At least 256GB SSD (Solid State Drive) to handle local blockchain node storage, dependencies, and temporary file streaming buffers.
- DISPLAY: Monitor with a resolution of 1366x768 or higher (Full HD 1920x1080 recommended for optimal dashboard rendering and transaction window monitoring).
- ADDITIONAL PERIPHERALS: Standard Keyboard, Mouse, and stable internet connectivity hardware for web3 node communication and live IPFS streaming.

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### **ii) SOFTWARE SPECIFICATIONS**

One of the most difficult tasks is selecting software. Once the system requirement is found out, we have to determine whether a particular software package fits those system requirements. This section summarizes the application requirement, assuming the system

uses a modern web development stack suitable for decentralized, data-intensive Web3 applications.

- OPERATING SYSTEM: Debian 12 / Windows 10+ / macOS Catalina or later.
- DEVELOPMENT ECOSYSTEM: Node.js runtime environment (v16.x or newer).
- PROGRAMMING LANGUAGES: Solidity (for decentralized ledger contracts) and JavaScript (ES6+ for frontend and backend logic).
- DATABASE (ON-CHAIN STATE): Ethereum Smart Contract Structs, Mappings, and Lists (acting as the immutable ledger database).
- DECENTRALIZED STORAGE: Pinata IPFS (InterPlanetary File System API Proxy).
- FRONT END TECHNOLOGIES: HTML5, CSS3, Bootstrap, Vanilla JavaScript, and Vue.
- CODE EDITOR / IDE: Visual Studio Code (VSCode).
- BLOCKCHAIN FRAMEWORK: Hardhat Development Environment (for contract compiling, local testing, and Sepolia deployment scripts).
- WEB3 PROVIDER / WALLET: MetaMask Browser Extension (for secure user authentication and transaction signing overrides).
- APIs / LIBRARIES: Ethers.js or Web3.js (for frontend-to-blockchain communication), Axios (for server-to-Pinata routing), and Multer (for handling multi-part file upload streams).

## **FEASIBILITY STUDY**

A comprehensive feasibility study was undertaken to evaluate the practicality and viability of developing the proposed intelligent e-learning platform. This study examined the project from three critical perspectives: technical feasibility, economic feasibility, and operational feasibility. Each dimension was carefully analyzed to ensure that the proposed system could be successfully implemented within available resources, technical capabilities, and organizational constraints while delivering sustainable value to end users. The feasibility study involved extensive research, data collection, and analysis across multiple dimensions to provide a holistic assessment of project viability. Market analysis was conducted to understand the current e-learning landscape, identify potential competitors, assess market demand, and evaluate growth opportunities. User surveys and interviews were performed with potential learners, educators, and educational administrators to gather requirements, understand pain points with existing solutions, and validate the proposed features. Technical experts were consulted to evaluate the suitability of the chosen technology stack, identify potential implementation challenges, and recommend best practices for system architecture

and development. Financial modeling was carried out to project development costs, operational expenses, revenue streams, and return on investment under various adoption scenarios. Risk assessment identified potential technical, financial, and operational risks along with mitigation strategies to ensure project success. Timeline analysis established realistic development phases, milestones, and resource allocation requirements. The study also considered regulatory requirements, data protection standards, and compliance needs across different geographical regions where the platform may operate. Stakeholder analysis evaluated the interests, influence, and expectations of all parties including learners, instructors, administrators, investors, and partners to ensure alignment and support. Infrastructure assessment examined hosting requirements, bandwidth needs, scalability considerations, and disaster recovery capabilities. The comprehensive nature of this feasibility study ensures that all aspects of project development and deployment have been thoroughly evaluated, providing confidence in the proposed system's viability and establishing a solid foundation for successful implementation.

#### **i) TECHNICAL FEASIBILITY**

The proposed system is technically feasible as it leverages mature, well-established technologies with extensive community support and proven reliability. The frontend will be developed using Flutter, Google's UI toolkit that enables building natively compiled applications for mobile, web, and desktop from a single codebase, ensuring consistent user experience across all devices while reducing development time and maintenance effort. The backend will be implemented using Node.js, a powerful JavaScript runtime built on Chrome's V8 engine, which offers exceptional performance for I/O-intensive operations, seamless handling of concurrent user requests, and a rich ecosystem of libraries through npm. MongoDB, a leading NoSQL database, will serve as the primary data store, providing flexible document-based storage that perfectly accommodates the dynamic and hierarchical nature of educational content, user profiles, scheduling data, and performance records. The integration of AI-powered examination and diagnostic analytics is technically achievable using machine learning frameworks and natural language processing libraries that can be integrated with the Node.js backend. Real-time virtual classroom functionality can be implemented using WebRTC technology or third-party APIs specializing in video conferencing. The technology stack is scalable, well-documented, and supported by large developer communities, ensuring that technical challenges can be addressed effectively during development. The required development expertise is readily available in the market, and the hardware infrastructure needed for hosting and deployment is accessible through cloud platforms at reasonable costs.

#### **ii) ECONOMIC FEASIBILITY**

The proposed system is economically feasible, with development and operational costs that are reasonable relative to the expected benefits and potential returns. The primary cost components include development resources, software licenses, cloud infrastructure, and



ongoing maintenance. Since the chosen technology stack consists of open-source technologies including Flutter, Node.js, and MongoDB, there are no significant licensing costs associated with the core development tools. Development costs are primarily driven by personnel expenses for skilled developers, UI/UX designers, and quality assurance testers, which can be optimized through careful project planning and resource allocation. Cloud hosting costs for servers, databases, and storage are predictable and scalable, allowing the platform to start with minimal infrastructure and expand as user adoption grows. Third-party services for video conferencing, if required, involve subscription costs that can be factored into the operational budget. On the revenue side, the platform can generate income through multiple streams including subscription plans for learners, commission from instructor sessions, premium features, and institutional partnerships. The system's ability to serve a global audience across diverse subject domains creates significant market potential. The elimination of manual processes, reduction in administrative overhead, and improved learning outcomes deliver substantial value to users, justifying the initial investment. A break-even analysis indicates that with moderate user adoption rates, the platform can recover development costs within a reasonable timeframe and generate sustainable returns thereafter.

### **iii) OPERATIONAL FEASIBILITY**

The proposed system is operationally feasible, as it aligns well with the needs and capabilities of its intended users while requiring manageable organizational changes for successful implementation. From the learner's perspective, the platform offers an intuitive interface with familiar navigation patterns, reducing the learning curve and encouraging adoption. Students can easily register, select their desired field from the hierarchical taxonomy, schedule sessions with preferred instructors, and access live virtual classrooms without specialized technical knowledge. The AI-driven instructor matching simplifies the process of finding suitable teachers, while automated attendance tracking eliminates manual effort. From the instructor's viewpoint, the platform provides clear visibility into scheduled sessions, student profiles, and performance analytics, enabling effective preparation and targeted instruction. Educators can focus on teaching rather than administrative tasks, enhancing their productivity and satisfaction. For platform administrators, the system offers centralized management of users, content, scheduling, and financial transactions through comprehensive dashboards and reporting tools. The operational processes including user registration, session scheduling, payment processing, and performance tracking are designed to be seamless and minimally disruptive to existing workflows. Training requirements for end users are minimal due to the intuitive design, while administrators and support staff can be quickly oriented through documentation and basic training sessions. The platform's scalability ensures that operational processes remain efficient as user numbers grow, and the availability of technical support ensures prompt resolution of any operational issues that may arise. Overall, the system integrates smoothly into the daily routines of all stakeholders, making its adoption and sustained operation highly feasible.

# SYSTEM DESIGN

## **USERS OF THE SYSTEM**

The LearNX is designed to cater to all types of users with distinct roles and functionalities that align with their operational responsibilities within the decentralized storage ecosystem.

### **i) ADMINISTRATORS (CONTRACT OWNERS)**

Administrators are the high-level system deployers responsible for the overall structural health, global parameter configuration, and operational control of the LearnX network environment. They connect via an authorized administrative Ethereum wallet address, granting them elevated smart contract permissions to execute critical global updates. Administrators can manage global storage parameters, adjust blockchain gas fee structural settings, configure base premium tier pricing values, and oversee system-wide contract operations. They are typically responsible for maintaining the underlying code integrity and auditing global storage trends.

### **ii) STANDARD USERS**

Standard users are the end-users who primarily interact with the storage platform for their day-to-day asset management and file sharing needs. They authenticate securely by connecting their individual cryptographic wallets through the MetaMask interface, granting them immediate access to their personalized cloud environment. Standard users focus on uploading files, managing private directory structures, monitoring their 250 MB storage tier limits, whitelisting peer addresses for secure file sharing, and interacting with the decentralized peer-to-peer asset marketplace. User functions are focused on execution, data privacy, and secure content ownership.

## **MODULARITY CRITERIA**

### **ADMINISTRATOR (CONTRACT OWNER)**

- - Connect Wallet / Secure Administrative Login – Administrator must securely authenticate using email-based login with multi-factor authentication or blockchain wallet connection to access the admin dashboard.
- - Configure Global Subscription and Storage Parameters – Administrator must have the ability to set and update global platform parameters including free tier storage limits (e.g., 250 MB), premium subscription pricing, and session duration caps for learners.

- - Manage Instructor Verification and Onboarding – Administrator must review, approve, or reject instructor registration requests, verify instructor credentials and subject matter expertise before granting teaching privileges.
- - Update Global Pricing Arrays for Subscription Plans – Administrator must be able to modify subscription plan prices, features, and validity periods across multiple learner tiers (Basic, Premium, Institutional).
- - Monitor Platform Operations and Live Sessions – Administrator must oversee real-time platform analytics including active users, ongoing live sessions, server load, and AI diagnostic engine performance.
- - Audit Global System State and User Activity Logs – Administrator must have read-only access to comprehensive audit logs including learner progress records, instructor performance metrics, attendance reports, and any policy violations.
- - Manage Announcements and System Notifications – Administrator must be able to broadcast platform-wide announcements, maintenance schedules, and policy updates to all learners and instructors.

### **STANDARD USER (LEARNER)**

- - Register / Secure Login – Learner must securely register and log in using email authentication, social login options, or blockchain wallet integration with role-based access enforcement.
- - Personalized Dashboard Access – Learner must access a consolidated dashboard displaying upcoming live sessions, completed session history, attendance records, AI-generated weakness reports, and personalized remedial recommendations.
- - Browse and Enroll in Live Sessions – Learner must be able to search, filter, and enroll in live sessions based on subject category, instructor expertise, schedule availability, and enrollment capacity.
- - Participate in Live Interactive Sessions – Learner must join scheduled live video sessions with real-time chat, screen sharing, and whiteboard capabilities, with automatic attendance tracking based on join time and engagement duration.
- - Access Structured Content Repository – Learner must be able to browse, search, and access pre-recorded video lectures, reading materials, quizzes, and assignments organized into subject folders, with progress tracking across completed materials.

- - View AI-Driven Diagnostic Reports – Learner must receive detailed diagnostic reports after each completed session, clearly identifying specific knowledge gaps, weak areas, and mastery levels with actionable improvement suggestions.
- - Access Remedial Content Recommendations – Learner must receive and access automatically generated personalized remedial learning paths including suggested video modules, practice quizzes, and one-on-one revision session requests.
- - Track Personal Learning Progress – Learner must monitor their own learning journey through visual progress charts showing session completion rates, assessment scores, time spent on remedial content, and overall subject mastery trends.
- - Manage Profile and Preferences – Learner must be able to update personal information, set learning preferences (preferred subjects, available time slots, learning pace), and manage notification settings.
- - View Immutable Learning History Logs – Learner must access a complete, append-only history of all completed sessions, attendance records, diagnostic reports, and remedial content accessed for personal audit and review.

### **STANDARD USER (INSTRUCTOR)**

- - Register / Secure Login – Instructor must securely register, submit credentials for verification, and log in after administrative approval with role-based teaching privileges.
- - Create and Schedule Live Sessions – Instructor must be able to create new live sessions by specifying subject, topic, description, schedule date and time, duration, maximum learner capacity, and required prerequisites.
- - Manage Session Content and Materials – Instructor must upload and organize pre-session reading materials, slides, quizzes, and assignments associated with each live session.
- - Host Live Interactive Sessions – Instructor must host live video sessions with real-time chat moderation, screen sharing, digital whiteboard, and the ability to record sessions for on-demand learner access.
- - Access Learner Attendance and Engagement Reports – Instructor must view detailed attendance lists for each session, including join times, participation duration, and engagement metrics for individual learners.

- - Review Learner Diagnostic Reports – Instructor must access aggregated diagnostic reports for their sessions to understand class-wide weak areas and common knowledge gaps, enabling targeted teaching improvements.
- - Provide Remedial Support – Instructor must receive and respond to learner requests for one-on-one revision sessions, and upload remedial materials addressing identified weak areas.
- - Upload Structured Content to Repository – Instructor must be able to upload, organize, edit, and remove pre-recorded video lectures, supplementary reading materials, practice quizzes, and assignments into categorized subject folders.
- - Monitor Instructor Performance Analytics – Instructor must access personal performance metrics including average session attendance, learner satisfaction ratings, diagnostic improvement trends across learners, and session completion rates.
- - View Teaching History Logs – Instructor must access a complete history of all past sessions taught, including attendance summaries, diagnostic outcomes, and learner feedback for continuous improvement.

# ARCHITECTURE DIAGRAMS

## DATA FLOW DIAGRAM

Data flow diagram is used to define the flow of the system and its resources such as information. As the name suggests DFD is a representation of the data flow within the system. It includes processes, data store and external interfaces to the system. External interfaces may be treated as external entities. DFD's can be exploded to sub-processes. Any two entities data store must be connected to each other by some process. The process cannot be directly connected to each other. The following are the four major components of the DFD:

- **Process**
- **Data store**
- **Data flow**
- **External Entity**

It is also known as the bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of the input data to system, various processing carried out in these data and output data generated by the system. The main reason why this DFD technique is so popular is probably because of facts that DFD is a very simple formalism. It is simple to understand and use. A DFD model uses a very limited number of primitive symbols to represent the functions performed by a system and the data flow among these systems. Starting with a set of high-level functions that a system performance of DFD model in hierarchically it represents various sub functions. The data flow diagramming technique also follows a simple set of intuitive concepts and rules.

### **Process**

Process shows the work of the system. Each process has one or more data inputs and produce one or more data outputs. Process is represented by rounded rectangles, circles or ovals in DFD's. Each process has a unique name and number. This name and number appear inside the rectangle that represents the process in a DFD.

### **Data store**

Data store is a repository of data. Processes can enter data into or retrieve the data from data store. Each data has a unique name

### **Data flow**

Data flow show the passage of data in the system and represented by lines joining system components. An arrow indicates the direction of flow and the line is labelled by name of the data flow.

### **External Entity**

External Entities are outside the system but they either supply input data into the system or use other system output. They are represented by rectangles. They are entities on which the designer has control. They may be any natural person, corporation, partnership, sole proprietorship, association, organization, company, government agency etc.

### **Level-0 DFD**

This is the simplest form of a DFD, representing the entire system as a single process. It shows how the system interacts with external entities like users or other systems. No details about data stores or internal processes are included, providing a high-level overview of data flow.

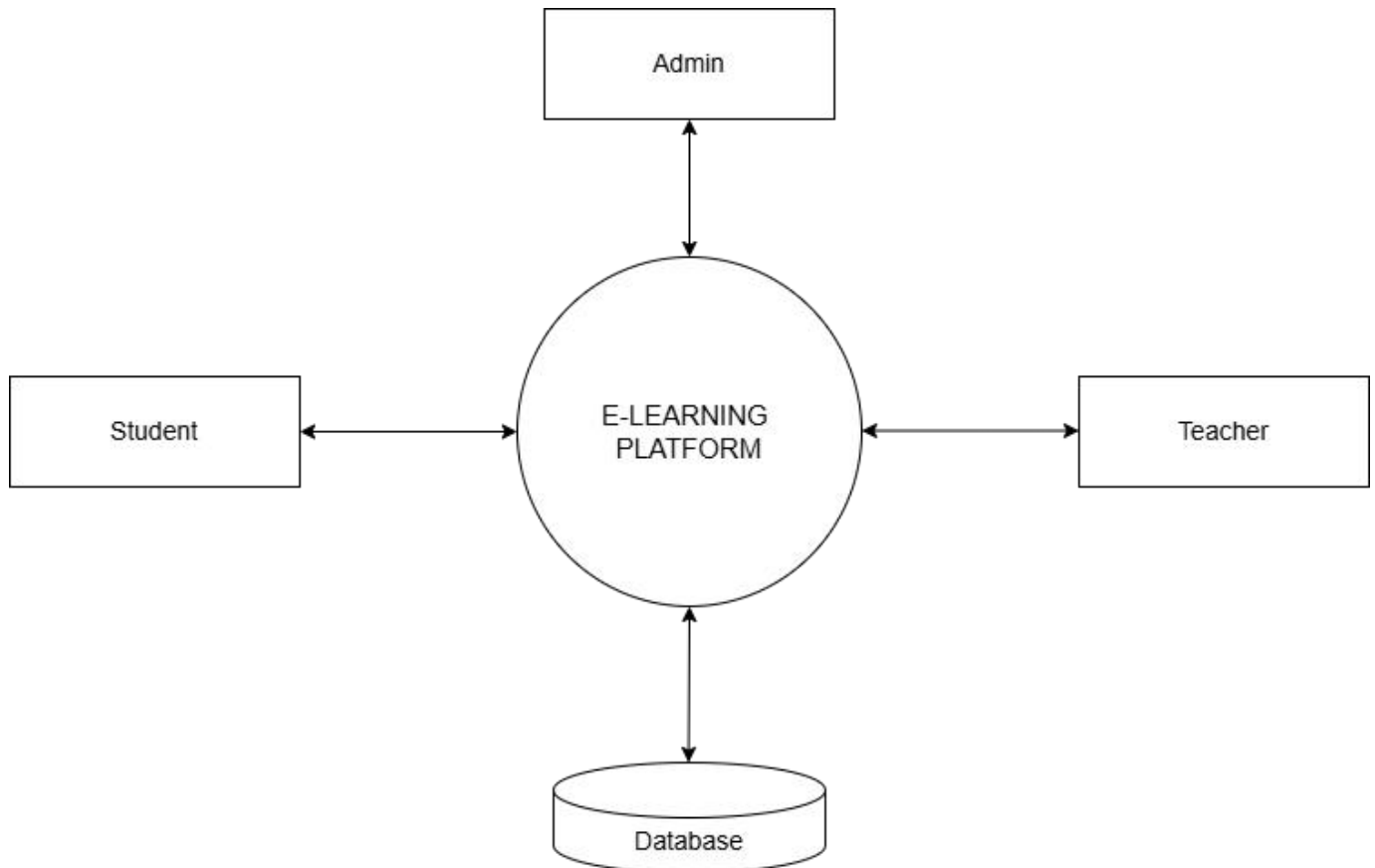
### **Level-1 DFD**

At this level, the system is broken down into its main processes to show how data flows between them. Data stores are introduced along with the flow of data between processes and external entities. This level provides a more detailed view of the system's core functionalities.

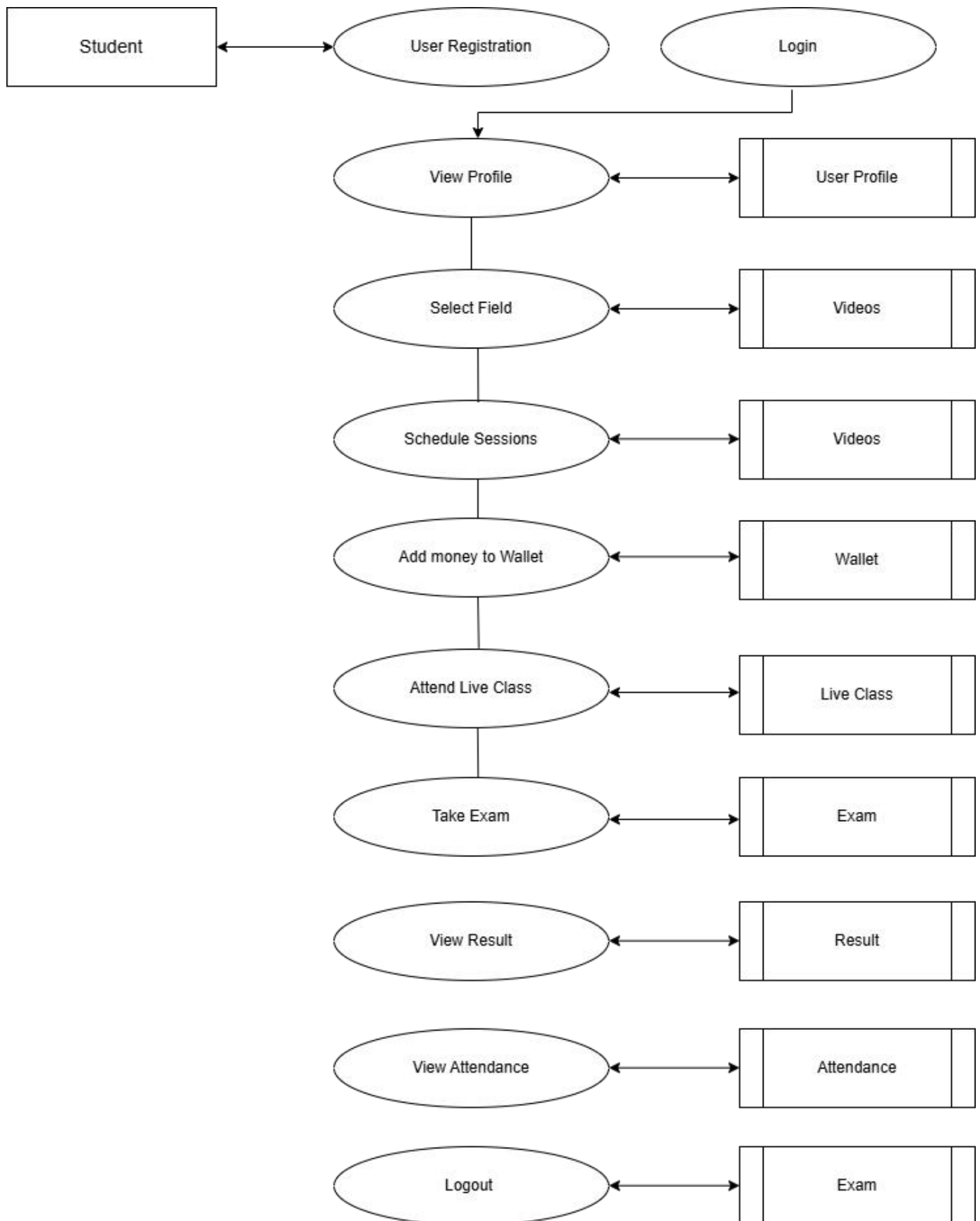


## DATA FLOW DIAGRAM

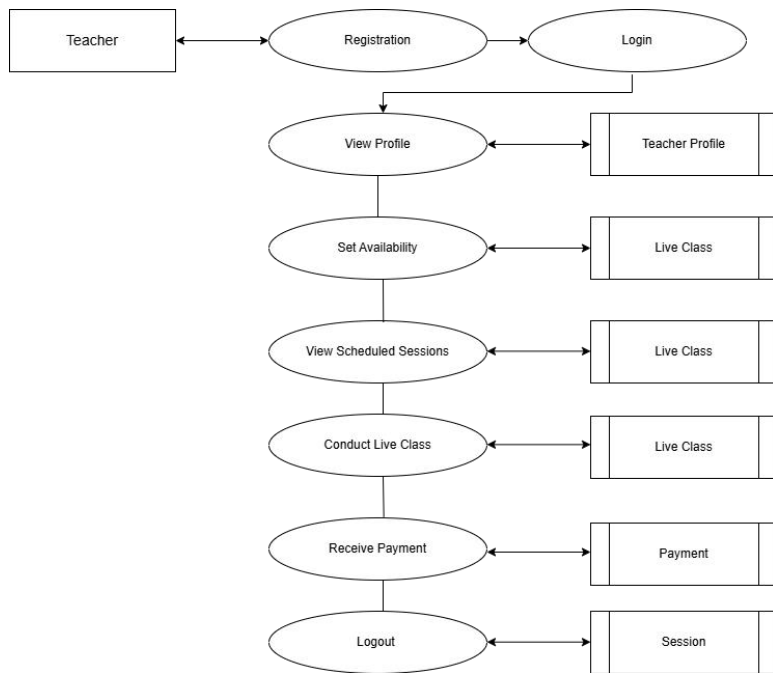
### DFD: Level 0



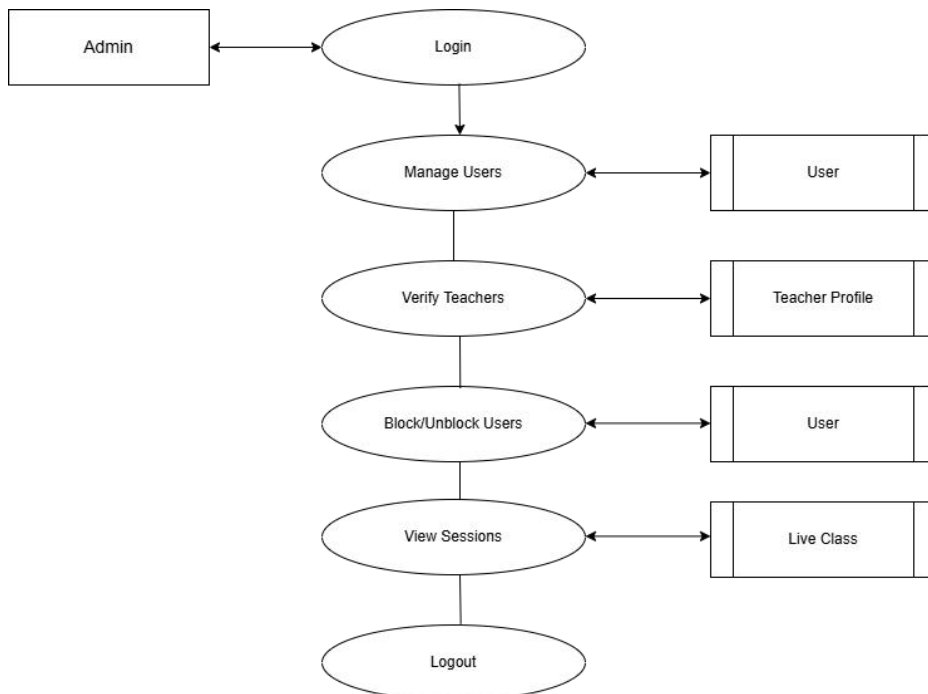
## DFD: Level 1



## DFD: Level 2



## DFD: Level 3



## Tables

**Table : Admin**

| Column Name | Data Type    | Constraints                  | Description                 |
|-------------|--------------|------------------------------|-----------------------------|
| admin_id    | INT          | PK, NOT NULL, AUTO INCREMENT | Unique identifier for admin |
| username    | VARCHAR(50)  | NOT NULL, UNIQUE             | Admin login name            |
| password    | VARCHAR(255) | NOT NULL                     | Encrypted password          |
| email       | VARCHAR(100) | NOT NULL, UNIQUE             | Admin email                 |
| created_at  | DATETIME     | DEFAULT CURRENT_TIMESTAMP    | Account creation date       |

## Table : User

| Column Name   | Data Type    | Constraints                  | Description                    |
|---------------|--------------|------------------------------|--------------------------------|
| user_id       | INT          | PK, NOT NULL, AUTO INCREMENT | Unique identifier for the user |
| full_name     | VARCHAR(100) | NOT NULL                     | User's full name               |
| username      | VARCHAR(50)  | NOT NULL, UNIQUE             | User login name                |
| password      | VARCHAR(255) | NOT NULL                     | Encrypted password             |
| email         | VARCHAR(100) | NOT NULL, UNIQUE             | User email address             |
| phone         | VARCHAR(20)  | NULL                         | User contact number            |
| registered_at | DATETIME     | DEFAULT CURRENT_TIMESTAMP    | Account creation date          |

**Table : Teacher**

| Column Name    | Data Type    | Constraints                      | Description                       |
|----------------|--------------|----------------------------------|-----------------------------------|
| teacher_id     | INT          | PK, NOT NULL, AUTO INCREMENT     | Unique identifier for the teacher |
| full_name      | VARCHAR(100) | NOT NULL                         | Teacher's full name               |
| email          | VARCHAR(100) | NOT NULL, UNIQUE                 | Teacher email address             |
| phone          | VARCHAR(20)  | NULL                             | Teacher contact number            |
| expertise_area | VARCHAR(255) | NULL                             | Main subjects they teach          |
| created_at     | DATETIME     | DEFAULT CURRENT_TIMESTAMP<br>AMP | Profile creation date             |

## Table : Exam

| Column Name | Data Type | Constraints                  | Description                                        |
|-------------|-----------|------------------------------|----------------------------------------------------|
| exam_id     | INT       | PK, NOT NULL, AUTO INCREMENT | Unique identifier                                  |
| session_id  | INT       | FK, NOT NULL, UNIQUE         | References Session(session_id) (One-to-One)        |
| exam_data   | JSON      | NULL                         | Stores questions and answers (can be parsed by AI) |
| created_at  | DATETIME  | DEFAULT CURRENT_TIMESTAMP    | Time exam was created                              |

**Table : Transaction**

| Column Name      | Data Type               | Constraints                  | Description                                     |
|------------------|-------------------------|------------------------------|-------------------------------------------------|
| transaction_id   | INT                     | PK, NOT NULL, AUTO INCREMENT | Unique identifier                               |
| wallet_id        | INT                     | FK, NOT NULL                 | References Wallet(wallet_id)                    |
| amount           | DECIMAL(10, 2)          | NOT NULL                     | Amount added or deducted                        |
| transaction_type | ENUM('Credit', 'Debit') | NOT NULL                     | Credit (adding money) or Debit (paying teacher) |



## Table : Session

| Column Name      | Data Type | Constraints                  | Description                                    |
|------------------|-----------|------------------------------|------------------------------------------------|
| session_id       | INT       | PK, NOT NULL, AUTO INCREMENT | Unique identifier for the class session        |
| course_id        | INT       | FK, NOT NULL                 | References Course(course_id)                   |
| teacher_id       | INT       | FK, NOT NULL                 | References Teacher(teacher_id) who is assigned |
| session_datetime | DATETIME  | NOT NULL                     | Date and time of the live class                |

Table : Wallet

| Column Name | Data Type      | Constraints                        | Description                                 |
|-------------|----------------|------------------------------------|---------------------------------------------|
| wallet_id   | INT            | PK, NOT NULL,<br>AUTO<br>INCREMENT | Unique identifier<br>for the wallet record  |
| user_id     | INT            | FK, NOT NULL,<br>UNIQUE            | References<br>User(user_id)<br>(One-to-One) |
| balance     | DECIMAL(10, 2) | NOT NULL,<br>DEFAULT 0.00          | Current amount in the<br>user's wallet      |

# SYSTEM DEVELOPMENT

## **SYSTEM DEVELOPMENT**

A software implementation method is integrating software-based services systematically. This is the phase in the software life cycle where the actual software is implemented. The result of this phase consists of source code, together with documentation to make the code more readable. Implementation is the stage of a project where the theoretical design is turned into a working system. If the implementation is not carefully planned and controlled, it may cause confusion.

The most crucial stage is achieving a successful system and giving users confidence that the system will work according to their requirements. It involves careful planning, investigation of the current system, and its constraints. The more complex the system, the more effort is needed to implement it. In other words, implementation is the process of bringing a developed system into operational use and turning it over to the user. Implementation activities extend from planning through converting the old system to the new one.

The system development of the LearnX AI-Driven Personalized E-Learning Platform was executed through an Agile and iterative process focused on rapid deployment, personalized learning experiences, and highly intuitive educational user interfaces. The initial planning phase involved comprehensive requirement gathering to define core learning needs like real-time instructor-learner matching, automated attendance tracking, accurate AI-driven diagnostic assessment, and personalized remedial content recommendation. The system design prioritized a scalable, web-based architecture utilizing a cloud-hosted database for user and progress management, a Node.js backend API for business logic orchestration, a third-party video streaming API for live sessions, and an AI microservice for intelligent diagnostics and reporting. Agile methodologies were employed to support continuous development, allowing the team to incrementally build, test, and refine modules such as the personalized learner dashboard, the intelligent session scheduling system, and the AI diagnostic reporting engine. Rigorous unit and integration tests were conducted to guarantee seamless data flow and correct diagnostic evaluations. The resulting system is a robust, intelligent, Web-based learning solution that delivers high efficiency, accurate personalized assessments, and seamless instructor-learner interaction.

### **1. ADMINISTRATOR DASHBOARD**

This module provides the central interface for the Platform Owner or Administrator, offering high-level operational and platform configuration oversight. It aggregates key data like total active learners count, active instructors count, ongoing live sessions, platform-wide attendance summaries, and AI diagnostic report generation statistics. It also provides tools for instructor verification and approval, global subscription plan configuration, and

system announcement broadcasting.

## **2. LEARNER AND INSTRUCTOR MANAGEMENT**

This module is responsible for the complete lifecycle of user accounts and role management. It includes tools for learner registration, instructor onboarding and verification, role-based access assignment, profile updates, and account status monitoring. It is the primary module for generating user capacity indicators, tracking active versus inactive users, and applying role-specific permission filters.

## **3. INTELLIGENT INSTRUCTOR-LEARNER MATCHING MODULE**

This module manages the intelligent matching system, including learner preferences, instructor expertise profiles, and matching histories. Crucially, it tracks the compatibility relationships between learners and instructors based on subject alignment, availability constraints, learning style compatibility, and past performance data, allowing users to view and receive automated instructor recommendations.

## **4. LIVE SESSION MANAGEMENT MODULE**

The core user interface for scheduling, hosting, and joining live interactive sessions. It features a rapid scheduling pipeline with integrated calendar view, real-time availability checking, and automatic attendance tracking functionality. It handles session creation, instructor assignment, learner enrollment, session recording capture, automatic join time logging, duration tracking, and final session completion reporting.

## **5. AI-DIAGNOSTIC ASSESSMENT MODULE**

This intelligence-focused module handles automated learner performance analysis. It calculates and generates detailed diagnostic reports after each completed session, identifying specific knowledge gaps, weak areas, and mastery levels. It automatically pushes personalized remedial content recommendations, including suggested video modules, practice quizzes, and one-on-one revision session requests, ensuring seamless continuous improvement.

## **6. PROGRESS AND ATTENDANCE TRACKING MODULE**

This background execution process analyzes learner session metrics against completion benchmarks and automatically pushes attendance records and progress updates to the dashboard interface. It tracks join times, participation duration, engagement metrics, session completion rates, assessment scores, and overall mastery trends, minimizing manual tracking efforts and ensuring accurate learning journey documentation.

## 7. CONTENT REPOSITORY & LEARNING HISTORY AUDIT

Focusing on absolute accountability, this module compiles comprehensive tracking records of all content accessed, sessions completed, diagnostic reports generated, and remedial materials consumed. It relies on immutable database properties to deliver a verifiable learning audit trail essential for learner verification and ensuring total system integrity

## 8. SOFTWARE DEVELOPMENT

- Frontend: Implemented using HTML5, CSS3, Bootstrap, React.js or Vanilla JavaScript to ensure highly responsive web navigation panels and a seamless user experience across devices.
- Architecture: The application operates as a single-page web application (SPA) with integrated video streaming APIs and real-time communication protocols for fast, frictionless learner-instructor interaction.
- Backend Integration: Server-side logic is achieved via a Node.js/Express API layer handling user authentication, session scheduling, attendance recording, AI diagnostic report generation, and content repository management.

## 9. DATABASE DESIGN

- Primary Database: A cloud-hosted relational database (such as PostgreSQL or MySQL) is utilized as the core persistent storage, providing structured tables for users, roles, sessions, attendance records, diagnostic reports, content metadata, and progress tracking, accessed via optimized API queries.
- File Storage Pipeline: The frontend and Node.js backend use API calls to route video recordings and pre-recorded content directly into cloud storage services (such as AWS S3 or similar), linking structural records back to the primary database using unique file identifiers and access URLs.

## 10. USER INTERFACE

- Intuitive and responsive UI design modeled after conventional cloud drives for fast folder and file interaction in the storage grid.
- Dashboard UI designed to clearly display storage tier KPIs, file category cards, and MetaMask connectivity badges.

## 11. TESTING AND EVALUATION

- Unit Testing: Executed for critical smart contract functions like gas limit optimizations, free tier constraint boundaries, and purchase allocation logic.
- Integration Testing: Conducted to ensure seamless data handoffs between the local Node.js proxy server, Pinata IPFS endpoints, and the MetaMask transaction window injection.

## **SYSTEM TESTING AND IMPLEMENTATION**

## **SYSTEM TESTING**

In general, testing is finding out how well software or a product works. Testing is a set of activity that can be planned and conducted systematically. Testing begins at the module level and works towards the integration of entire computer-based systems. Nothing is complete without testing, as it is vital to the success of the system.

Software testing is a process of executing a program or application with the intent of finding the software bugs. Testing is for evaluating a system or its components to find whether it satisfies the specified requirements or not. Testing cannot show the absence of defects; it can also be stated as the process of validating and verifying a software program, system, or product.

## **TESTING TECHNIQUES**

- Unit Testing
- Integration Testing
- Validation Testing
- System Testing
- Output Testing
- User Acceptance Testing

## **UNIT TESTING**

It is the first level of testing. Each module is tested individually, and focus is given to finding errors limited to each individual module and correcting them. The different modules of the system are tested individually and corrected for all errors. Each module is focused on working satisfactorily with regard to the expected output from the module.

Here, we verify the proper functioning of individual components like the user login authentication hook, live session scheduling logic, attendance tracking calculation, and AI diagnostic report generation engine. For Example: Testing the instructor-learner matching algorithm to ensure that trying to match a learner with an incompatible instructor based on subject misalignment or availability conflict accurately prevents the match and provides appropriate alternative recommendations.

## **INTEGRATION TESTING**

Integration testing is a systematic test for constructing the program structure while at the same time conducting tests to uncover errors. Instead of testing the system as a whole, it focuses on how the modules that make up the system interact. It tests for the errors resulting



from the integration of modules. One specific target of integration testing is the interface between modules. The objective is to take unit-tested modules and build a program structure dictated by design.

Here, we check whether modules like the Live Session Management, Intelligent Instructor-Learner Matching, and the AI Diagnostic Assessment engine work together as expected. For Example: Ensuring that a successful live session completion accurately triggers the AI diagnostic engine to analyze learner performance, generates a detailed weakness report, automatically updates the learner dashboard with attendance records, and pushes personalized remedial content recommendations to the learner's profile.

### **VALIDATION TESTING**

Validation testing is performed to ensure the system meets the functional requirements specified by the user. For instance, checking the Contract Administrator dashboard to ensure role-based settings modifications are correctly granted or denied based on the connected public key. Ensure all critical features, such as Decentralized Marketplace operations and user file sorting filters, operate correctly.

For Example: Checking that executing a file purchase transaction within the Marketplace module accurately calculates token amounts in Wei, transfers owner rights on the blockchain state ledger, updates access variables, and instantly shifts the bought asset into the recipient's "My Files" inventory.

### **SYSTEM TESTING**

System testing was performed to verify that all system elements have been properly integrated and perform their allocated function within the complete system environment. Security testing was done to check the security mechanisms built into the system, which will protect it from improper penetration (e.g., ensuring unwhitelisted Ethereum addresses cannot decrypt or view private IPFS hashes). Performance testing was done to test the runtime performance of the software. For user acceptance testing, the system was given to the end-user to use. Confirm that the entire app operates seamlessly, including both Administrator and Standard User functionalities.

For Example: Ensuring that the web application handles multiple concurrent standard user uploads and heavy IPFS payload lookups across diverse browsers using MetaMask extensions without stalling the interface or encountering Web3 RPC data concurrency errors.

## **OUTPUT TESTING**

After performing the validation testing, the next step is the output testing of the enhanced system. No system could be useful if it does not produce the required output in the required format. The outputs generated or displayed by the system are tested by asking the users about the format required by them. Check that outputs like the file history log table, storage capacity progress cards, and marketplace directory displays present accurate business and ledger data based on user inputs.

For Example: Ensuring that the generated on-chain history record displays the correct transaction hash string, the precise byte size, the corresponding asset file name, and an accurate Unix block timestamp of the event.

## **USER ACCEPTANCE TESTING**

User acceptance testing comprises a completed and successful end-to-end system test with review of the results by one or more users with specific knowledge. Users may apply a variety of validation techniques. For example: generating a transaction report from the new system and comparing the results with the same ledger state from the test network explorer, data inspection, and others. Prior to this testing, the system was delivered to the clients along with the steps for implementation.

In user acceptance testing, the system was run in the client environment by system users. The users were allowed to test the system and raise any issues for a specific period after the system is assumed to be complete and error-free. Here, we collect feedback from Administrators and Users about the application's functionality and usability.

For Example: Allowing a test user to connect their MetaMask wallet, upload an asset payload using the user dashboard interface, designate a second test wallet to a file access whitelist mapping, and have that second user independently verify successful download resolution of the pinned document to ensure all decentralized actions work seamlessly for storage operations.

## **SYSTEM IMPLEMENTATION**

A software implementation method is integrating software-based services systematically. This is the phase in the software life cycle where the actual software is implemented. The result of this phase consists of source code, together with documentation to make the code more readable. Implementation is the stage of a project where the theoretical design is turned into a working system. If the implementation is not carefully planned and controlled, it may cause confusion.

Implementation is done only after testing is complete. The most crucial stage is achieving a new successful system and giving confidence about the new system to the user that it will work according to the requirements. It involves careful planning, investigation of the

current system, and its constraints on implementation. The more complex the system is, the more effort is needed to implement the system. In other words, implementation is the process of bringing a developed system into operational use and turning it over to the user. Implementation activities extend from planning through the old system to the new one.

The implementation involves the following formalities:

- **Careful planning:** Formulating structured milestones for compiling the Solidity source contracts via Hardhat, environment variable synchronization for Pinata API endpoints, and main deployment procedures.
- **Investigation of the systems and constraints:** Evaluating browser Web3 environments and handling gas fee parameters on the Sepolia network test layers to minimize user transaction failures.
- **Design the methods to achieve the changes:** Structuring proxy backend server gateways and Ethers.js integration listeners to securely transition user asset workflows from legacy local structures to a decentralized paradigm.
- **Training the staffs in the changed phase:** Providing instructional guides to users regarding MetaMask setup instructions, secure private key preservation rules, and transaction confirmation flows.
- **Evaluation of the changes over method:** Reviewing live storage performance metrics, gas optimizations, and repository consistency records against initial user privacy and sovereign expectations.

# **SYSTEM MAINTENANCE**

## **SYSTEM MAINTENANCE**

A process of modifying a software system or component after deployment to correct faults, improve performance, and adapt to changing requirements is known as software maintenance. Software maintenance is not limited to fixing defects; it also includes enhancing existing features, optimizing system performance, removing obsolete functionalities, and ensuring compatibility with evolving technologies. Since changes in software environments are inevitable, maintenance activities involve evaluating, controlling, and implementing modifications throughout the system's operational life. The primary objective of maintenance is to preserve and enhance the value of the software over time by improving usability, efficiency, security, and functionality. While software development may take only a few years, maintenance can continue for a much longer period to ensure the system remains reliable and relevant.

For the LEARNX AI-Driven Personalized E-Learning Platform , corrective maintenance is performed to resolve faults and errors encountered during daily operation, such as live video streaming interruptions, AI diagnostic engine processing failures, real-time chat synchronization issues, attendance tracking inaccuracies, and database query errors that affect learner progress retrieval and dashboard rendering.

Adaptive maintenance is carried out to accommodate changes in external technologies and services, including updates to third-party video conferencing APIs (such as Zoom, Daily, or LiveKit), new versions of WebRTC protocols for real-time communication, changes in cloud storage service providers for recorded session archiving, modifications to AI model deployment frameworks, and evolving browser security policies affecting WebRTC and media capture functionalities.

Perfective maintenance focuses on improving system performance and user experience by optimizing the instructor-learner matching algorithm for faster and more accurate recommendations, reducing AI diagnostic report generation latency, enhancing dashboard responsiveness across devices, refining the user interface for better accessibility and navigation, improving video streaming quality under varying network conditions, and implementing smarter remedial content recommendation logic based on longitudinal learner performance data.

Preventive maintenance is undertaken to reduce the possibility of future failures by conducting regular security audits of user data storage, identifying potential vulnerabilities in authentication mechanisms, monitoring server load and auto-scaling thresholds, implementing automated health checks for all critical services (video streaming, AI engine, database, notification system), performing database backups and integrity verification, stress-testing the platform under high concurrency scenarios, and regularly updating dependencies to patch known security vulnerabilities.

# CONCLUSION

## **CONCLUSION**

In conclusion, LEARNX addresses the fundamental shortcomings of traditional online learning platforms by offering an integrated, intelligent, and learner-centric ecosystem. By combining structured content delivery, real-time live interaction, intelligent instructor matching, and AI-driven diagnostic assessments, the platform effectively eliminates the inefficiencies of manual coordination and surface-level evaluation. It empowers learners across all categories to identify and bridge their knowledge gaps systematically while providing a consolidated view of their attendance and progress. The automation of critical processes such as weakness analysis and remediation recommendations ensures that learning becomes more personalized, efficient, and outcome-driven. Ultimately, LEARNX transforms the educational journey from a fragmented, time-consuming experience into a seamless, adaptive, and mastery-focused process, fulfilling the growing demand for intelligent solutions in the rapidly evolving landscape of digital education.

## **SCOPE FOR FUTURE ENHANCEMENT**

Future enhancements will significantly expand LEARNX's capabilities beyond core personalized learning, live interaction, and AI-driven assessment. A key focus is introducing **\*\*Blockchain-Based Credentialing and Certification\*\***, enabling learners to receive tamper-proof, verifiable digital certificates and micro-credentials upon successful completion of courses or mastery of specific subjects. These credentials can be shared directly with employers or academic institutions, eliminating manual verification processes.

The system can be scaled to support Multi-Institution and Enterprise Learning Repositories, allowing schools, universities, and corporate training departments to create their own segregated learning environments under a single platform instance. This includes centralized administration, cross-institution analytics, and shared instructor pools while maintaining data isolation and privacy compliance.

Further upgrades include seamless integration with **Advanced Adaptive Learning Algorithms\*\*** powered by reinforcement learning, which will dynamically adjust remedial content difficulty and pacing in real-time based on continuous learner performance monitoring. This moves beyond static diagnostic reports to truly personalized, self-optimizing learning pathways.

The addition of Gamification and Peer Learning Engines will boost learner engagement and retention through experience points (XP), achievement badges, leaderboards, and structured peer-to-peer study groups with collaborative problem-solving sessions. Learners will be able to form virtual study circles, share notes, and conduct peer reviews under instructor moderation.

Integration of Natural Language Processing (NLP) for Intelligent Chatbots will provide learners with 24/7 academic support, answering subject-related queries, explaining concepts, and guiding them through remedial content without requiring live instructor intervention. This significantly reduces response times for common questions and enhances the self-service learning experience.

Finally, integration with Learning Tools Interoperability (LTI) Standards will allow LEARNX to connect with existing Learning Management Systems (LMS) such as Moodle, Canvas, and Blackboard, enabling institutions to adopt LEARNX's AI-driven diagnostic and live session capabilities without abandoning their current infrastructure. This ensures the system evolves into a comprehensive, interoperable, and future-proof intelligent learning ecosystem.



# **BIBLIOGRAPHY**

## **BIBLIOGRAPHY**

### **WEBSITES**

- <http://chatgpt.openai.com>
- <https://docs.ipfs.tech/>
- <https://docs.soliditylang.org/>
- <https://docs.pinata.cloud/>
- <https://docs.ethers.org/v6/>

### **BOOKS**

- Online education : foundations ,planning and pedagogy
- Digital learning design:Designing Effective Online Learning Design
- Small Teaching online:Applying Learning Science in online Classes

# APPENDIX

## SOURCE CODES

### SAMPLE

```
6  class ApiService {
12  Future<void> register(
23      'role': role,
24      'categories': categories,
25  ),
26      headers: {'Content-Type': 'application/json'},
27  );
28  if (res.statusCode != 200) throw Exception(res.body);
29  final body = jsonDecode(res.body) as Map<String, dynamic>;
30  currentEmail = email;
31  currentRole = role;
32  currentCategories = List<String>.from(body['user']['categories'] ?? []);
33  }
34
35  Future<void> login(String email, String password) async {
36      final res = await http.post(
37          Uri.parse('$baseUrl/auth/login'),
38          body: jsonEncode({'email': email, 'password': password}),
39          headers: {'Content-Type': 'application/json'},
40      );
41      if (res.statusCode != 200) throw Exception(res.body);
42      final body = jsonDecode(res.body) as Map<String, dynamic>;
43      currentEmail = email;
44      currentRole = body['user']['role'] as String?;
45      currentCategories = List<String>.from(body['user']['categories'] ?? []);
46  }
47
48  static void logout() {
49      currentEmail = null;
50      currentRole = null;
51      currentCategories = [];
52  }
53
54  Future<List> fetchVideos({List<String>? types}) async {
55      final params = <String, String>{};
56      if (types != null && types.isNotEmpty) {
```

```

body: SingleChildScrollView(
  padding: const EdgeInsets.all(16.0),
  child: Form(
    key: _formKey,
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.stretch,
      children: [
        TextFormField(
          decoration: const InputDecoration(
            labelText: 'Email',
            border: OutlineInputBorder(),
          ), // InputDecoration
          validator: (v) =>
            (v == null || v.isEmpty) ? 'Enter email' : null,
          onSave: (v) => email = v ?? '',
        ), // TextFormField
        const SizedBox(height: 16),
        TextFormField(
          decoration: const InputDecoration(
            labelText: 'Password',
            border: OutlineInputBorder(),
          ), // InputDecoration
          obscureText: true,
          controller: _passwordController,
          validator: (v) =>
            (v == null || v.isEmpty) ? 'Enter password' : null,
        ), // TextFormField
      ],
    ),
  ),
)

```

```

}

class _TeacherHomePageState extends State<TeacherHomePage>
  with SingleTickerProviderStateMixin {
  final _formKey = GlobalKey<FormState>();
  late TabController _tabController;
  String title = '';
  String url = '';
  File? selectedFile;
  List myVideos = [];

  @override
  void initState() {
    super.initState();
    _tabController = TabController(length: 2, vsync: this);
    _loadMyVideos();
  }

  @override
  void dispose() {
    _tabController.dispose();
    super.dispose();
  }

  Future<void> _loadMyVideos() async {
    final api = ApiService();
  }
}

```

## SCREENSHOTS

